

Task 1 — Document 09: Technology Stack Specification & Justification

Problem Statement ID: PS 26117 / SIH26117 **Title:** Sovereign On-Premise Agentic AI Workbench using Open-Weight Multimodal LLMs for Confidential Industrial Work **Organization:** Mangalore Refinery and Petrochemicals Limited (MRPL) **Category:** Software | **Theme:** Smart Automation

■ ■ Technology Stack Summary

The **Sovereign AI Workbench** is built using modern open-source software, high-efficiency open-weight quantized models, and standard web technologies.

Layer	Component	Chosen Technology	Version / Spec	Purpose / Function
Frontend UI	Client Interface	Vanilla HTML5 / CSS3 / JS	Standard ES6+	Lightweight, zero-dependency dark-themed SPA
Backend Gateway	Web Server	FastAPI + Uvicorn	FastAPI 0.115+	High-async Python backend framework & API router
Model Runtime	LLM Runtime Daemon	Ollama	Latest (v0.5+)	Local GGUF quantized model hosting & inference
General / RAG LLM	Core Language Model	qwen3.5:4b	4B parameters (q4_K_M)	Reasoning, general conversation, RAG answer synthesis
Coding LLM	Specialized Coder	qwen2.5-coder:7b	7B parameters (q4_K_M)	Technical code generation and debugging
OCR Model	Multimodal OCR	glm-ocr:q8_0	Q8_0 quantization	Text extraction from images & scanned logs
Embeddings Model	Vector Generator	nomic-embed-text	768 dimensions	High-speed semantic vector embeddings
Vector Search Engine	Vector Store	In-Memory Cosine Similarity	Pure Python (math/numpy)	Vector matching with zero external DB overhead
PDF Processing	Document Parser	pypdf	5.0+	Extractor for digital text PDFs
Environment	Python Runtime	Python	3.14.6	Execution environment

■ Tech Stack Justification & Open-Weight Model Selection

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■ OPEN-WEIGHT MODEL SELECTION RATIONALE ■
■ qwen3.5:4b ■ Ultra-fast inference on standard workstation CPU, ■
■ vision-capable, excellent reasoning for its size. ■
■ qwen2.5-coder:7b ■ Benchmark-leading open code generation model, ■
■ handles Python, SQL, C++, bash scripts natively. ■
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[illegible]

Why Open-Weight Models over Cloud APIs?

- 1. 100% Data Sovereignty:** Confidential MRPL refinery SOPs and inspection logs never leave internal servers.
- 2. Air-Gap Compliance:** Zero external API dependencies; robust against cloud downtime or internet outages.
- 3. Zero Recurring API Costs:** Eliminates per-token cloud API billing.

■ Hardware & On-Premise Infrastructure Requirements

The workbench is optimized to run efficiently on standard workstation laptops or single-node edge servers without requiring expensive GPU server clusters.

Minimum Hardware Specification (Development / Single-User Workstation)

- **CPU:** Intel Core i7 / AMD Ryzen 7 (8 cores / 16 threads)
- **RAM:** 16 GB DDR4 / DDR5
- **Storage:** 50 GB available SSD space (for Ollama model GGUF weights)
- **OS:** Windows 11 / Linux (Ubuntu 22.04 LTS / RHEL 9)
- **GPU:** Optional (CPU execution fully supported)

Recommended Production Hardware Specification (Plant Server / Multi-Engineer Edge Node)

- **CPU:** Intel Xeon / AMD EPYC (16+ cores)
- **RAM:** 32 GB – 64 GB DDR5
- **GPU:** 1x NVIDIA RTX 4090 / RTX 4080 (16GB - 24GB VRAM) or NVIDIA A10G
- **Storage:** 100 GB NVMe M.2 SSD
- **Network:** Local 1 Gbps / 10 Gbps Air-Gapped LAN (No WAN access needed)

■ Latency & Benchmark Metrics

Operation	Model / Tool	Execution Time (CPU)	Execution Time (GPU)
Text Embedding (500 chars)	<code>nomic-embed-text</code>	~ 50 ms	~ 10 ms
Grounded RAG Query	<code>qwen3.5:4b</code>	1.5 – 3.0 s	0.4 – 0.8 s
Image OCR Extraction	<code>glm-ocr:q8_0</code>	3.0 – 6.0 s	0.8 – 1.5 s
Code Generation	<code>qwen2.5-coder:7b</code>	4.0 – 8.0 s	1.0 – 2.0 s